Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CS3DS01]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER-2021

INTRODUCTION TO DATA SCIENCE

Time:3 Hours **Maximum Marks:70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	a.	Define Data Science and Business Analytics. Explain the differences between Data Science and Business Analytics.	09
	b.	Explain the challenges and different opportunities in Business Analytics.	05
2.	a.	What is statistical inference? Explain the usage of Statistical inference in Data Science.	07
	b.	Explain the different applications of statistical techniques in Data Science with suitable examples.	07
3.	a.	What is Data Visualization? Explain the different applications of Data Visualization techniques.	07
	b.	Why machine learning is useful in Data Science? Explain the major differences of Supervised machine learning and Unsupervised machine learning.	07
		OR	
4.	a.	Discuss the role of SQL and NoSQL in the Data Science	07
	b.	Explain the different business intelligence tools and its crucial role in Data Science project presentation.	07
5.	a.	Explain the different challenges and scope of Data Science project management.	07
	b.	Write a brief note on Data Science project framework and execution flow of a Data Science project.	07
		OR	
6.	a.	Identify and explain about the different components of the Data Science project.	07
	b.	Write a note on the different Data Science project methods.	07
7.	a.	Explain the differences between exploratory data analysis and descriptive data analysis techniques using examples.	07
	b.	What is the role of Data Warehousing in Data Science? Explain the steps involved in the data warehousing process.	07
		OR	
8.	a.	Write a note on the various steps involved in the data warehouse construction.	07
	b.	What is the Role of Business Intelligence tool in the data Science project Presentation? Explain with an example.	07